Cat Detection using YOLOv8s

Objective

The objective of this project is to develop a deep learning based object detection model capable of detecting cats in images using the YOLOv8 architecture. The trained model identifies cats, generates bounding boxes around detected objects, and provides confidence scores for each prediction.

Technologies Used

- Python
- PyTorch
- YOLOv8s (Ultralytics)
- OpenCV
- Google Colab (Tesla T4 GPU)

Environment Setup

```
!pip install ultralytics
!pip install roboflow
```

```
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotli
```

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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>
Downloading roboflow-1.3.8-py3-none-any.whl (207 kB)
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Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Installing collected packages: pillow-avif-plugin, filetype, pi-heif, opencv-python-headless, idna, robofl
  Attempting uninstall: opencv-python-headless
```

Dataset Extraction and Preparation

```
import zipfile

zip_ref = zipfile.ZipFile('/content/cat_dataset.zip', 'r')
zip_ref.extractall('/content/dataset')
zip_ref.close()
```

Dataset Verification

```
import os

os.listdir('/content/dataset')
```

```
['data.yaml',
 'train',
 'README.roboflow.txt',
 'valid',
 'test',
 'README.dataset.txt']
```

Data Visualization

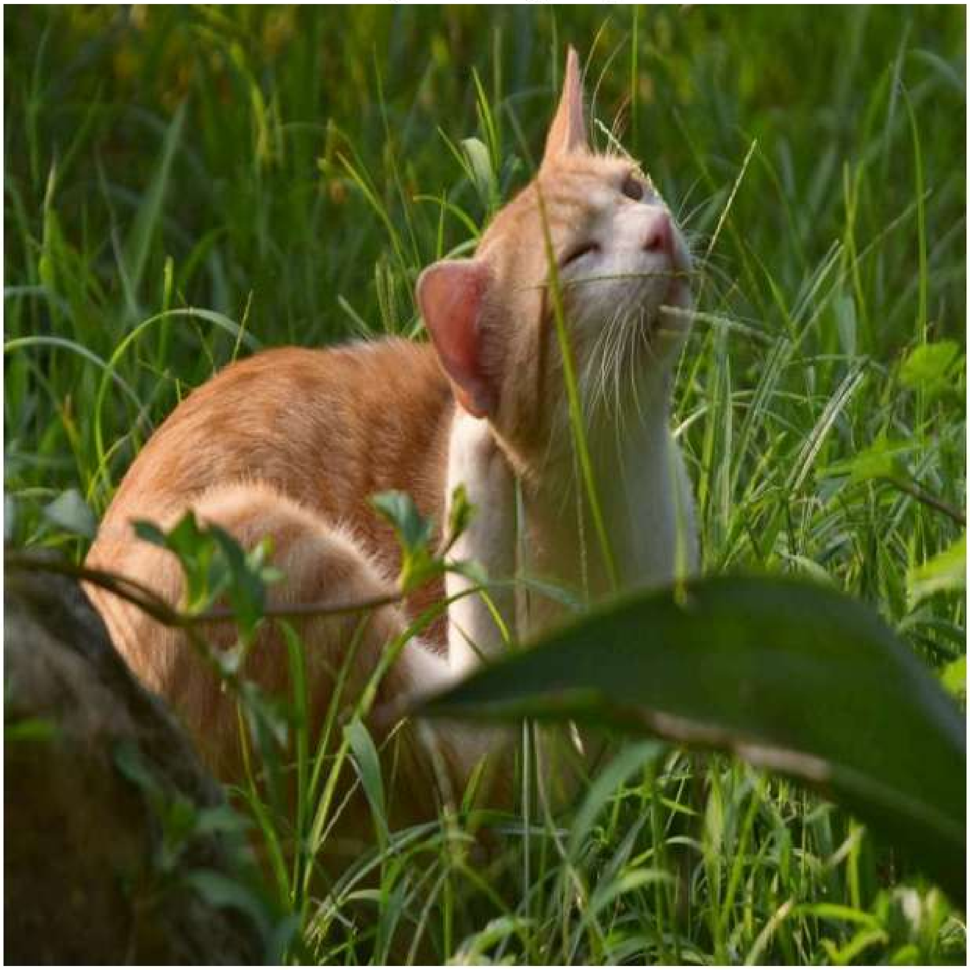
```
import cv2
import matplotlib.pyplot as plt
import glob

image_path = glob.glob('/content/dataset/train/images/*.jpg')[0]

image = cv2.imread(image_path)
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,8))
plt.imshow(image)
plt.axis("off")
plt.title("Sample Training Image")
plt.show()
```

Sample Training Image



Model Initialization

The YOLOv8s pretrained model is loaded using transfer learning.

```
from ultralytics import YOLO

model = YOLO("yolov8s.pt")
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see htt
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8s.pt to 'yolov8s.pt': 100%
```

Model Training

The model is trained on the custom cat detection dataset using YOLOv8s.

Training Configuration

- Epochs: 100
- Image Size: 640
- Batch Size: 8
- Optimizer: SGD

Data Augmentation Techniques

The following augmentation techniques were applied to improve model generalization:

- Horizontal flipping
- Mild rotation
- Translation
- Scaling
- HSV color augmentation
- Mosaic augmentation

```
results = model.train(

    data='/content/dataset/data.yaml',

    # Training
    epochs=100,
    imgsz=640,
    batch=8,
    device=0,

    # Optimizer
    optimizer='SGD',
    lr0=0.001,

    # Early stopping
    patience=30,

    # LIGHT AUGMENTATION
    degrees=5.0,
    translate=0.05,
    scale=0.1,

    fliplr=0.5,
    flipud=0.0,

    hsv_h=0.01,
    hsv_s=0.3,
    hsv_v=0.2,

    mosaic=0.5,
    mixup=0.0,

    # Validation
    val=True,
    save=True
)
```

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50       0.74        0.7      0.663      0.348

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     52/100       2.4G     0.8455     0.6415      1.334          3        640: 100% ━━━━━━━━━━━━ 18/18 5.4
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50      0.721      0.673      0.659      0.339

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     53/100       2.4G     0.8979     0.5871      1.323          3        640: 100% ━━━━━━━━━━━━ 18/18 6.6
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50      0.712       0.68      0.656      0.346

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     54/100       2.4G      0.844     0.5905      1.315          3        640: 100% ━━━━━━━━━━━━ 18/18 6.4
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50      0.733       0.68      0.676      0.356

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     55/100       2.4G     0.8901     0.6837      1.364          3        640: 100% ━━━━━━━━━━━━ 18/18 4.5
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50      0.762       0.72      0.697       0.36

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     56/100       2.4G      0.834     0.5899       1.33          3        640: 100% ━━━━━━━━━━━━ 18/18 6.4
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50      0.748       0.72      0.692      0.357
EarlyStopping: Training stopped early as no improvement observed in last 30 epochs. Best results observed
To update EarlyStopping(patience=30) pass a new patience value, i.e. `patience=300` or use `patience=0` to

56 epochs completed in 0.061 hours.
Optimizer stripped from /content/runs/detect/train/weights/last.pt, 22.5MB
Optimizer stripped from /content/runs/detect/train/weights/best.pt, 22.5MB

Validating /content/runs/detect/train/weights/best.pt...
Ultralytics 8.4.52 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 11,125,971 parameters, 0 gradients, 28.4 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━
                   all         43         50      0.736       0.74      0.794      0.456
Speed: 0.4ms preprocess, 4.5ms inference, 0.0ms loss, 4.1ms postprocess per image
Results saved to /content/runs/detect/train
```

Model Evaluation

The trained model is evaluated using standard object detection metrics:

- Precision
- Recall
- mAP50
- mAP50-95

```
metrics = model.val()
```

```
Ultralytics 8.4.52 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 11,125,971 parameters, 0 gradients, 28.4 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1468.9±325.8 MB/s, size: 55.5 KB)
val: Scanning /content/dataset/valid/labels.cache... 43 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━
                   all         43         50      0.737       0.74      0.792      0.456
Speed: 14.9ms preprocess, 14.7ms inference, 0.0ms loss, 3.4ms postprocess per image
Results saved to /content/runs/detect/val
```

Inference on Test Images

The trained model is tested on unseen images from the test dataset.

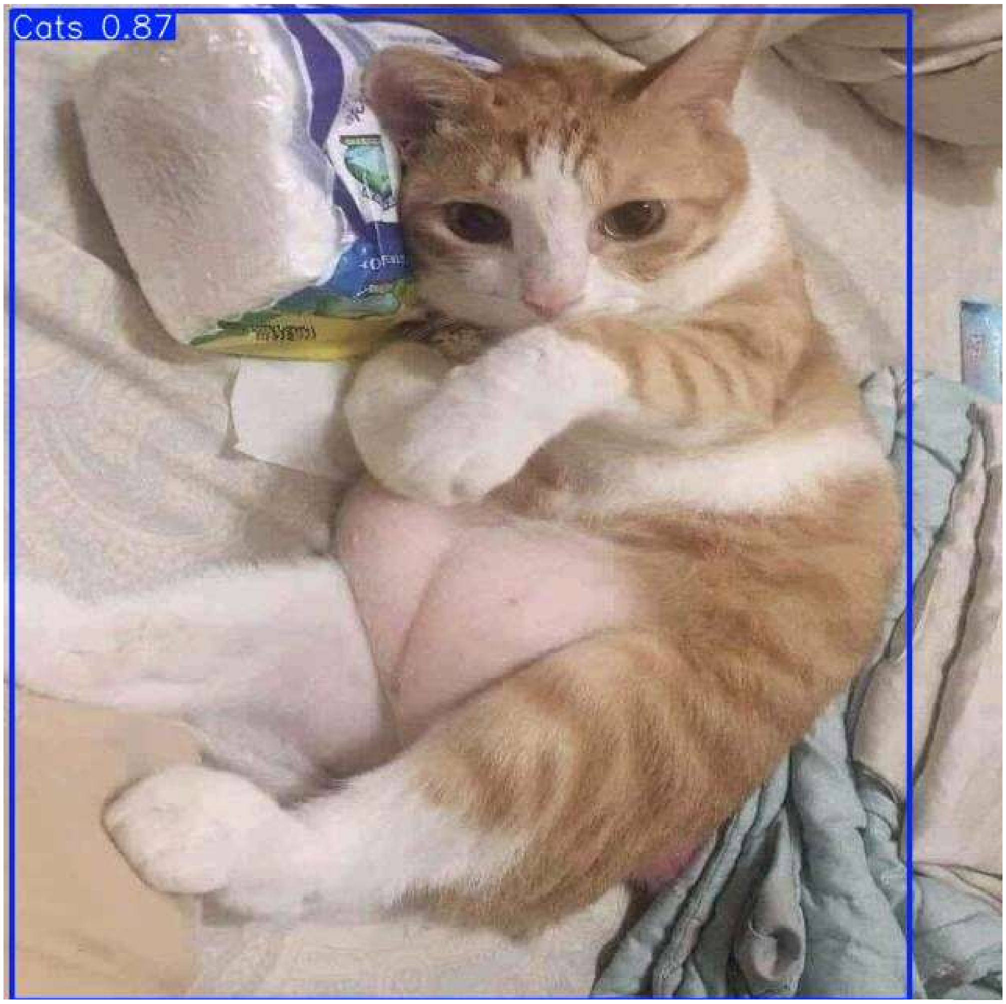
```
results = model.predict(
    source='/content/dataset/test/images',
    save=True,
    conf=0.25
)
```

```
image 1/18 /content/dataset/test/images/0066-J3l4Zf33nfc_jpg.rf.29cf8bb11494237408203b179fbc2e30.jpg: 640x6
image 2/18 /content/dataset/test/images/0262-jo9XwI6B8Rs_jpg.rf.0dfad9458ae98a798c411706ec0d4818.jpg: 640x6
image 3/18 /content/dataset/test/images/0340-LfHN48aCVtE_jpg.rf.b29d11a34cd60ddb9ccd4e5942ce7425.jpg: 640x6
image 4/18 /content/dataset/test/images/0366-6B-WeK_fT8M_jpg.rf.a2c564572c501418b8384f52f8b2dbf3.jpg: 640x6
image 5/18 /content/dataset/test/images/0502-xxdTZFljIfw_jpg.rf.408cc1f5131c04170e06126fd6c94337.jpg: 640x6
image 6/18 /content/dataset/test/images/1101-WMH1xzEznuY_jpg.rf.6afdc30101bc2166c923ead59c30c8c8.jpg: 640x6
image 7/18 /content/dataset/test/images/1377-QvqfyLHM6GM_jpg.rf.e5a0cbbb84d902298c63f2063e71b8b3.jpg: 640x6
image 8/18 /content/dataset/test/images/1501-22bl-hths7o_jpg.rf.fe41675d0dc955d02ca0addf08897841.jpg: 640x6
image 9/18 /content/dataset/test/images/1590-66NrNlC36u8_jpg.rf.ec514de2752f67fc18981cc56a29794a.jpg: 640x6
image 10/18 /content/dataset/test/images/1614-7xSWybUM7l0_jpg.rf.dec1150b6c798bfd24aa083610b4dbb0.jpg: 640x
image 11/18 /content/dataset/test/images/1718-MqXWogtgq18_jpg.rf.087255a53e5af76216ab43215a6af8ca.jpg: 640x
image 12/18 /content/dataset/test/images/1981-VZm3NEcemiE_jpg.rf.05b5017051e95845f19b717e2580a1af.jpg: 640x
image 13/18 /content/dataset/test/images/2T7OkT9RmTY_jpg.rf.0bc6b6738872e4869ac6613c5bb71446.jpg: 640x640 1
image 14/18 /content/dataset/test/images/3136-m4ojA8PCiSU_jpg.rf.472daa57f59191b8e16af78e3379d686.jpg: 640x
image 15/18 /content/dataset/test/images/3pA7x0G8xOY_jpg.rf.fe4e973214e201aefaa7768cb3c60594.jpg: 640x640 1
image 16/18 /content/dataset/test/images/Ic64CrKUNJY_jpg.rf.c64df192d7a434e3e4e24816ed257100.jpg: 640x640 (
image 17/18 /content/dataset/test/images/S2IGnPjVCJs_jpg.rf.8cdab7d3101ec5a1395808b6cdc848bb.jpg: 640x640 2
image 18/18 /content/dataset/test/images/m8Y1ELIEJ60_jpg.rf.f16d7485658107658f53bcfd8a49da9a.jpg: 640x640 1
Speed: 1.8ms preprocess, 13.5ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 640)
Results saved to /content/runs/detect/predict
```

```
from IPython.display import Image, display
import glob

predicted_images = glob.glob('/content/runs/detect/predict/*.jpg')

display(Image(filename=predicted_images[0]))
```



Inference on Custom Images

Testing on Non-Cat Images

Additional experiments were performed using images containing:

- humans
- dogs

```
from ultralytics import YOLO

model = YOLO('/content/runs/detect/train/weights/best.pt')
```

```
results = model.predict(
    source='/content/3.jpg',
    conf=0.5,
    save=True
)
```

```
image 1/1 /content/3.jpg: 480x640 (no detections), 48.7ms
Speed: 2.1ms preprocess, 48.7ms inference, 0.6ms postprocess per image at shape (1, 3, 480, 640)
Results saved to /content/runs/detect/predict-3
```

```
from IPython.display import Image, display

display(Image(filename='/content/runs/detect/predict-3/3.jpg'))
```

```
from ultralytics import YOLO
model = YOLO('/content/runs/detect/train/weights/best.pt')

results = model.predict(
    source='/content/2.jpg',
    conf=0.75,
    save=True
)
```

```
image 1/1 /content/2.jpg: 384x640 1 Cats, 11.3ms
Speed: 1.9ms preprocess, 11.3ms inference, 1.2ms postprocess per image at shape (1, 3, 384, 640)
Results saved to /content/runs/detect/predict-5
```

```
from IPython.display import Image, display

display(Image(filename='/content/runs/detect/predict-5/2.jpg'))
```


```
from ultralytics import YOLO
model = YOLO('/content/runs/detect/train/weights/best.pt')

results = model.predict(
    source='/content/4.jpg',
    conf=0.75,
    save=True
)
```

```
image 1/1 /content/4.jpg: 480x640 (no detections), 14.2ms
Speed: 6.9ms preprocess, 14.2ms inference, 0.9ms postprocess per image at shape (1, 3, 480, 640)
Results saved to /content/runs/detect/predict-6
```

```
from IPython.display import Image, display

display(Image(filename='/content/runs/detect/predict-6/4.jpg'))
```

Observations and Findings